

Source:

https://kb.ettus.com/index.php?action=pdfbook&format=single&title=TwinRX_Getting_Started_Guides

Note: Direct terminal download was blocked by Cloudflare HTTP 403, so this local PDF was regenerated from the official Ettus KB content.

Generated UTC: 2026-04-28T11:27:17.039017+00:00

Contents

1. Kit Contents
2. USRP Compatibility
3. Hardware Setup
4. Local Oscillator Connectors
5. Software Setup
6. Test and Verify the Operation of the USRP
7. Proper Care and Handling
8. Technical Support and Community Knowledge Base
9. Legal Considerations
10. Sales and Ordering Support
11. Terms and Conditions of Sale

Kit Contents

- TwinRX Daughterboard
- Two front panel cables: 1.4 inch SMA bulkhead female to MMCX right-angle male
- Two front panel washer and bolts
- Two LO sharing cables: 5 inch MMCX right-angle male to MMCX right-angle male
- Screw kit
- Hardware installation guide
- Getting started guide

USRP Compatibility

- USRP X300/X310

Hardware Setup

For complete step-by-step instructions to install a daughterboard, see the TwinRX Module Installation Guide or the USRP X Series Quick Start daughterboard installation guide.

Local Oscillator Connectors

LO sharing cables are not required for a single TwinRX setup. LO sharing cables are required with two TwinRX daughterboards in a single USRP X300/X310.

Connector table

- J1: LO2 Export, min 0 dBm, nominal 0 dBm, output only
- J2: LO2 Input, min 0 dBm, damage +20 dBm

J3: LO1 Export, min -12 dBm, nominal +5 dBm, output only
J4: LO1 Input, min -10 dBm, nominal -5 dBm, damage +10 dBm

LO Sharing with Neighbour TwinRXs

TwinRX A slot J1 LO2 Export -> TwinRX B slot J2 LO2 Input
TwinRX A slot J2 LO2 Input <- TwinRX B slot J1 LO2 Export
TwinRX A slot J3 LO1 Export -> TwinRX B slot J4 LO1 Input
TwinRX A slot J4 LO1 Input <- TwinRX B slot J3 LO1 Export

Software Setup

- UHD minimum version: 3.10.0.0
- GNU Radio minimum version: 3.7.10

Install and configure the UHD/GNU Radio toolchain before using the TwinRX. Use a current stable UHD where possible.

Test and Verify Operation

The UHD driver includes example programs that can verify receiver operation.

Example RX1 command:

```
/usr/local/lib/uhd/examples/rx_ascii_art_dft --freq 88.1e6 --rate 1e6 --gain 70 --ref-lvl  
-30 --subdev "A:0" --ant RX1
```

To test RX2, change the antenna flag to RX2 and the subdevice spec to A:1.
Type Q or Ctrl-C to stop rx_ascii_art_dft.

GNU Radio GUI example:

```
uhd_fft -s 1e6 -g 70 -f 88.1e6 --spec "A:0" -A RX1
```

Proper Care and Handling

- Never allow metal objects to touch the circuit board while powered.
- Always handle the board with proper anti-static methods.
- Never allow the board to directly or indirectly contact voltage spikes.
- Never allow water or condensing moisture to contact the boards.
- Always use caution with FPGA, firmware, or software modifications.
- Never apply more than +10 dBm of power into any RF input.
- Always use at least 30 dB attenuation if operating in loopback configuration.

Support and Legal

Technical support: support@ettus.com

Community mailing list: usrp-users

Users are responsible for complying with applicable radio laws and regulations.

Terms and conditions: <http://www.ettus.com/legal/terms-and-conditions-of-sale>